

Risk Management

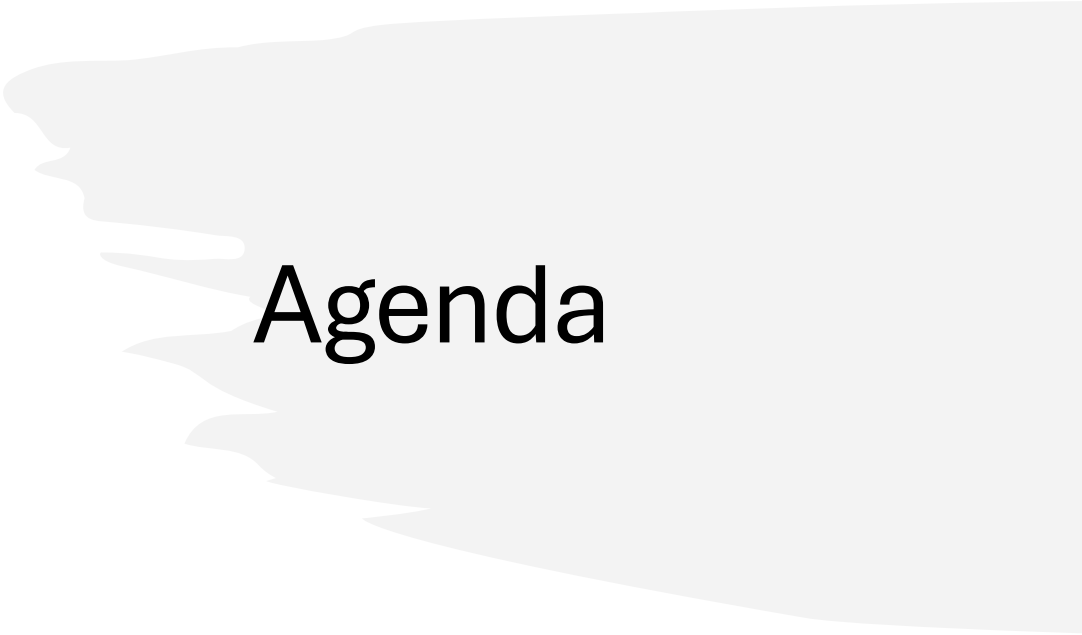
CIS 4032 – TOPIC 5

Mr. Madusanka Nipunajith

Intended Learning Outcomes

- Identify potential risks in software projects
- Categorize and prioritize risks
- Calculate the RE of risks
- Calculate RRL for countermeasures
- Make plans to deal with risks

- AOA Technique (Another Day)
- Risk Management
 - What is Risk
 - What is RM
 - A framework for dealing with risks
 - Identification
 - Estimation
 - Planning
 - Monitoring
 - Evaluate the effort of uncertainty
 - PERT
 - Exercises



Agenda

What is a risk?

- An unforeseen/unexpected event may or may not occur in the future that can adversely impact the project's cost, schedule, or quality.
- Relates to the future (Risk is future-oriented).
- Measured in terms of probability of occurrences and consequences.
- Involves cause and effect.
 - Every risk has a cause (why it might happen) and a potential effect (what might happen if it does).
- 2 types of risks
 - Positive Risks (Opportunities)
 - Negative Risks (Threats)

Risk Management (RM)



An attempt to minimize the chances of failure caused by unplanned events/risks.



Aim to identify, quantify, and take actions to minimize the effects of risks on the project.



Risk management involves additional costs.



Involves 2 key components

Risk Assessment

Risk Control

Risk Assessment + Risk Control



Risk assessment is about understanding the risks before deciding what to do about them.



Think of it as answering:

What can go wrong?
How likely is it to happen?
If it happens, how serious will the impact be?



Activities in Risk Assessment

Risk Identification
Risk Analysis
Risk Prioritization

Risk Assessment + Risk Control



After assessing the risks, the project team decides what actions to take.



Think of it as answering:

Now that we know the risks,
how do we manage them?



Activities in Risk Control

Risk Planning (4 Activities)
Risk Monitoring

A framework for dealing with risks



Risk Identification



Risk estimation (Analysis and
Prioritization)



Risk planning



Risk monitoring



Risk Identification

- Two main approaches
 - Checklist
 - List a set of risks that have been found more likely to occur and suggest countermeasures for each risk.
 - Predefined list based on PPs.
 - Brainstorming (encourage free thinking)
 - Representatives of the main stakeholders discuss and identify the problems that might occur.

Risk Estimation

		Impact →				
		Negligible	Minor	Moderate	Significant	Severe
Likelihood ↑	Very Likely	Low Med	Medium	Med Hi	High	High
	Likely	Low	Low Med	Medium	Med Hi	High
	Possible	Low	Low Med	Medium	Med Hi	Med Hi
	Unlikely	Low	Low Med	Low Med	Medium	Med Hi
	Very Unlikely	Low	Low	Low Med	Medium	Medium

- Risk Exposure
 - Expected value of the loss for the risk.
- Risk Likelihood
 - The probability of the risk
- Risk Impact
 - The total loss incurred if the risk happens.

$$\text{Risk} = \text{Likelihood} \times \text{Impact}$$

Likelihood = f (Vulnerabilities, Threats, Exposure, Security Controls)

Impact = g (Business Criticality)



Example

- A project depends on a data center, which is vulnerable to fire. It is estimated that there is a 1 in 1000 (1/1000) chance fire happening. If a fire occurred, a new computer configuration could be established for Rs 500,000. Find the risk exposure.
- $RE = 500,000 \times 0.001 = \text{Rs } 500$
- This is the minimum sum an insurance company would require as a premium.

Calculating Risk Exposure

Risk Exposure, RE

Risk Exposure calculates the effective current cost of a risk, and can be used to prioritize risks that require countermeasures.

Risk Exposure, RE = Probability of risk occurring × Total loss if risk occurs

$$RE = 2\% \times 80K = \frac{2}{100} \times 80000 = 1600$$

Risk	Probability of occurring	Total loss if it occurs	Risk Exposure
Product recall situation	2%	80K	1600
Significant product rejection	0.1%	1000K	1000
Competitive strike	10%	25K	2500

Highest Risk Exposure indicates most serious risk

Exercise - University Food Delivery App

Risk	Probability (%)	Impact (\$)
A. Server crashes during peak hours	25%	5000
B. Competitor launches similar app	60%	15000
C. Key developer quits mid-project	30%	8000
D. Food safety incident with partner restaurant	5%	50000
E. University bans food delivery on campus	10%	40000
F. The payment system gets hacked	15%	25000
G. App store rejects our app	20%	3000

1. Calculate Risk Exposure for each risk
2. Rank the risks from highest to lowest exposure
3. Identify the most serious risk and explain why

Answer - University Food Delivery App

Risk	Probability (%)	Impact (\$)	RE
A. Server crashes during peak hours	25%	5000	1250
B. Competitor launches similar app	60%	15000	9000
C. Key developer quits mid-project	30%	8000	2400
D. Food safety incident with partner restaurant	5%	50000	2500
E. University bans food delivery on campus	10%	40000	4000
F. The payment system gets hacked	15%	25000	3750
G. App store rejects our app	20%	3000	600

1. Calculate Risk Exposure for each risk
2. Rank the risks from highest to lowest exposure
3. Identify the most serious risk and explain why

Answer - University Food Delivery App

- Risk Ranking (Highest to Lowest):
- B. Competitor launches - \$9,000 - **Most Serious Risk**
- E. University bans delivery - \$4,000
- F. Payment hack - \$3,750
- D. Food safety incident - \$2,500
- C. Developer quits - \$2,400
- A. Server crashes - \$1,250
- H. App store rejection - \$600

Risk Planning

- How to deal with risks?
 - Risk Acceptance (Tolerate)
 - Do nothing. Live with the risk because it's small, or fixing it costs too much.
 - Example: Not buying phone insurance for an old phone.
 - Risk Avoidance (Terminate)
 - Don't do the risky activity at all.
 - Example: Not driving to avoid car accidents.
 - Risk Reduction (Treat)
 - Make the risk smaller or less harmful.
 - Example: Wearing a helmet when biking.
 - RRL
 - Risk Transfer
 - Transfer risk to another party
 - Insurance, Outsourcing

Risk Monitoring

- Deciding on risk reduction action
 - Some risks can be reduced or avoided immediately.
 - 4th phase of the framework.
 - Risk Reduction Leverage (RRL)
 - $RRL = (RE_{before} - RE_{after}) / RRC$
 - RRC – Risk Reduction Cost
 - Identify the best countermeasure out of multiple countermeasures for mitigating the risk

Risk Reduction Leverage

- Gives the ROI of the “countermeasures” being considered for mitigating risk
- Computed as:

$$\frac{\text{Reduction in Risk}}{\text{Cost of Countermeasure}} = \frac{RE_{before} - RE_{after}}{\text{Risk Reduction Cost}}$$

- Useful for prioritizing various mitigation strategies

Example: Use of RRL

Risk Reduction Leverage, RRL

Risk Reduction Leverage calculates a value for the 'return on investment' for a countermeasure and can thus be used to prioritize possible countermeasures.

$$\text{Risk Reduction Leverage of a countermeasure, RRL} = \frac{\text{Reduction in Risk Exposure}}{\text{Cost of the countermeasure}}$$

$$\text{RRL} = \frac{2500 - 150}{40000} = 0.059$$

Risk		Probability of occurring	Total loss if it occurs	Risk Exposure	
Competitive strike		10%	25K	2500	
Countermeasure	Total cost	New risk probability	New total loss	New RE	RRL
Advertising campaign	40K	3%	5K	150	0.059
Price promotions	30K	5%	10K	500	0.067
Simultaneous launch	10K	8%	15K	1200	0.13

Highest RRL indicates most cost-effective countermeasure

Exercise

Risk	Probability of occurring	Total loss if it occurs	Risk Exposure		
	15%	800K	120K		
System breach					
Countermeasure	Total cost	New risk probability	New total loss	New RE	RRL
Enhanced security training	20K	8%	64K		
Advanced intrusion detection	50K	5%	40K		
Complete system overhaul	80K	3%	24K		

Highest RRL indicates most cost-effective countermeasure

Answer

Risk	Probability of occurring	Total loss if it occurs	Risk Exposure		
	15%	800K	120K		
System breach					
Countermeasure	Total cost	New risk probability	New total loss	New RE	RRL
Enhanced security training	20K	8%	64K	5120	5.744
Advanced intrusion detection	50K	5%	40K	2000	2.360
Complete system overhaul	80K	3%	24K	720	1.488

Highest RRL indicates most cost-effective countermeasure

Evaluate the effects of uncertainty

- Techniques to consider the uncertainties in the duration of activities within a project.
- PERT (Program Evaluation Review Technique)
 - $E = (O + 4M + P) / 6$
 - $S = (P - O) / 6$
 - E – Expected Duration
 - O – Optimistic Time
 - P – Pessimistic Time
 - M – Most Likely Time
 - S – Standard Deviation
- Monte Carlo / Probability simulation
- Critical Chain Management

PERT Formula:

$$E = (O + 4 \times M + P) / 6$$

E= Estimate

O= Optimistic

M= Most likely

P= Pessimistic

Evaluate the effects of uncertainty

- These values are NOT calculated mathematically at the start.
- They are:
 - Collected from experts
 - Based on historical data
 - Or team discussion/judgment
- So PERT is a probabilistic estimation method, not a fixed formula system.
- Once O, M, and P are given, we calculate - Expected time (TE)
 - This gives a weighted average, where:
 - Most likely (M) has the highest weight
 - Pessimistic and optimistic contribute less

PERT Formula:

$$E = (O + 4 \times M + P)/6$$

E= Estimate

O= Optimistic

M= Most likely

P= Pessimistic

Example - PERT



Writing a Research Paper

Optimistic (O): 3 days (if you have all sources ready)

Most Likely (M): 7 days (normal research and writing)

Pessimistic (P): 15 days (if you get stuck or need major revisions)



Calculations:

Expected Duration: $E = (3 + 4 \times 7 + 15) / 6 = 46/6 = 7.7$ days

Risk Level: $S = (15 - 3) / 6 = 2$ days

What this means: Plan for 8 days, but know it could vary by ± 2 days (uncertainty).

Exercise

Activity	O (wk)	M (wk)	P (wk)	E (wk)	S (wk)
A	5	6	8	6.17	0.5
B	3	4	5	?	?
C	2	3	3	2.83	0.17
D	3.5	4	5	?	?
E	1	3	4	2.83	0.5
F	8	10	15	?	?
G	2	3	4	?	?
H	2	2	2.5	2.08	0.08

Exercise - Answer

Activity	O (wk)	M (wk)	P (wk)	E (wk)	S (wk)
A	5	6	8	6.17	0.5
B	3	4	5	4.00	0.33
C	2	3	3	2.83	0.17
D	3.5	4	5	4.08	0.25
E	1	3	4	2.83	0.5
F	8	10	15	10.50	1.17
G	2	3	4	3.00	0.33
H	2	2	2.5	2.08	0.08

PERT CHART

PERT Chart and AON are closely related, but they are not the same thing.

PERT

- Answers - How long might each activity take under uncertainty?
- Uses optimistic, most likely, and pessimistic time estimates
- Focuses on uncertainty in activity durations
- Can be drawn as a network diagram (AOA)

AON

- Answers - In what order should activities be performed, and when can they start and finish?
- Usually uses a single duration estimate
- Focuses on activity relationships and scheduling
- Is itself a network diagram

PERT CHART – Historical Difference

- Originally
 - PERT used Activity-on-Arrow (AOA) networks.
 - Modern project management software (MS Project, Primavera, etc.) typically uses Activity-on-Node (AON) networks.
- Therefore
 - Many textbooks today combine the ideas and say:
 - "Draw a PERT chart."
 - Drawing an **AON network** with PERT time estimates.
- PERT is the technique used to estimate uncertain activity durations, while AON is the network diagram used to represent activities and their dependencies. In practice, PERT calculations are often performed on an AON diagram.

Pros & Cons of PERT

- Pros
 - Better Time Estimation
 - PERT uses three time estimates (optimistic, pessimistic, and most likely), giving a more realistic and flexible schedule.
 - Identifies Critical Path
 - Helps managers identify the critical path, showing the task sequence that directly affects project duration.
 - Helps in Risk Management
 - By considering uncertainties in time estimates, risks become visible before project execution, and potential delays can be addressed proactively.
 - Supports Decision-Making
 - Provides useful data for what-if analysis and scenario planning in uncertain environments.

Pros & Cons of PERT

- Cons
 - Time-Consuming to Create
 - Developing a detailed **PERT chart** with multiple time estimates for each activity can be complex and slow.
 - Requires Accurate Data
 - PERT relies on subjective estimates. Inaccurate inputs can lead to misleading results.
 - Not Ideal for Short Projects
 - For small or routine projects, PERT might be overkill, adding unnecessary complexity.
 - Difficult to Update
 - Once a project starts, updating the PERT chart as changes happen can be difficult and time-consuming.

Exercise/Assignment

Activity	Dependency	T(o)	T(m)	T(p)	T(e)	T(s)
A		8	10	12		
B	A	5	7	9		
C	A	1	2	3		
D	B	6	8	10		
E	C,D	2	3	5		
F	-	11	13	17		
G	E,F	4	6	8		
H	G	9	11	14		

1. Identify each activity's values for T(e) and T(s).
2. Draw the AON diagram.
3. Calculate the Total Float of each activity.
4. Calculate the Free Float of each activity.
5. Identify the critical path.



Next Lecture

- Topic 6
 - Resource Allocation
 - What is resource allocation?
 - Result of resource allocation?
 - Activity Schedule
 - Resource Schedule
 - Cost Schedule
 - Crashing a project
 - Exercises



Thank you